

YI DONG

Embedded Linux / Embedded Systems / SoC Intern

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Education

Imperial College London

London, UK

MEng Electronic and Information Engineering

Expected graduation: 2028

- Relevant coursework: Digital Electronics, Computer Architecture, Signals and Systems, Communications, Control Systems, Embedded Systems, Mathematics for Engineers, Software Systems.

Skills

Embedded / Linux: C/C++, Python, Linux, Bash, runtime integration, socket communication, OpenCV, device bring-up, debugging

MCU / Devices: ESP32, ESP8266, GPIO/PWM, sensors, motors, OLED, basic serial communication, peripheral integration

SoC / FPGA Integration: PYNQ-Z1, Zynq-7000, PS/PL integration, AXI DMA usage, MMIO result access, Vivado basics

Tools: Git, Makefile basics, Verilator, Icarus Verilog, Vivado/XSim, waveform debugging

Projects

FPGA Wand / PYNQ-Z1 Vision System

Feb 2026–Mar 2026

Linux/Python, OpenCV, UDP sockets, PYNQ-Z1, AXI DMA, MMIO, PS/PL integration

GitHub

- Coordinated OpenCV camera capture, DMA dataflow, FPGA overlay management, and UDP point streaming in a Linux/Python runtime for a PYNQ-Z1 vision prototype.
- Offloaded pixel-level threshold statistics and centroid accumulation to FPGA logic, using MMIO readback to reduce PS-side per-frame processing work.
- Brought up the system on real hardware; debugged camera initialization, DMA transfer behavior, overlay/HWH mismatches, repeated frames, and PS/PL handoff issues.

Mini-Shell / Linux Systems Programming

Nov 2025

C, Linux, fork/exec, pipes, I/O redirection, command parsing

GitHub

- Implemented a simplified Linux shell with command parsing, fork/exec, cd, pipes, and I/O redirection, strengthening understanding of the Unix process model.
- Debugged child-process reaping, file-descriptor handling, nested command parsing, and a small raw-mode line editor with tab completion.

UDP Multithreaded Chat / Network Programming Exploration

Initial exploration

UDP socket exploration, client/server design, threading concepts

GitHub

- Built a UDP socket communication prototype to explore client/server message flow, send/receive thread separation, and basic network-program error handling.

Smart Balcony Embedded Exploration

Personal exploration

ESP32, sensors, motors, GPIO/PWM

- Explored ESP32-based automation prototypes with sensors and simple actuators, gaining hands-on practice with GPIO/PWM control, peripheral integration, sensor acquisition, and device debugging.

RV32I CPU / Architecture and Systems Understanding

Nov 2025–Dec 2025

SystemVerilog, RV32I, Verilator, waveform debugging

GitHub

- Implemented and tested a simplified RV32I pipeline, using Verilator, directed assembly, and waveforms to understand datapath/control flow, hazards, writeback, and memory/cache interactions.